

Practical Assignment [2026-09-22]

1. Find data with a dependent variable(s) that is not necessarily continuous
2. For the practical assignment I would like you to run at least 4 GLMs. Your report should be neat and have consistent formatting (font, font size, alignment, *etc.*).

[The task is prove to me that you can do GLMs. It doesn't help just copying and running code off the internet]

You must:

- Run a naïve linear model (Gaussian/Normal OLS) that will serve as a comparison for your GLMs.
- You should then run appropriate GLMs, with different linear predictors. Run several different models – Changing the random component/link function, keeping the linear predictor the same.
- [If you remove or add additional predictors in the first draft, you needn't TEST the additional predictors; this will only be part of the assignment *after* the first interview].

I want to see the following for the assignment:

- Rationale for the model (3 components) you are choosing (in each GLM)
- A very brief explanation of any data problems and how you solved these.
- The code for the models you ran.
- The model outputs.
- Interpretation of the slope parameter estimates
- You can include comparison of AICs (since you have done these in your other module). Remember that AICs for models with different Y spaces can't be compared).
- **Mention if and why a model might not be appropriate for your data, as well as any major differences in the interpreted parameters across models.**

Rubric

Marks:	Absent	Partial	Mostly there	Perfect
Model explanations	0	1	2	3
Data discussion	0	1	1	2
Code	0	1-2	3-4	5
Outputs	0	1-2	3-4	5
Interpretation (and major changes across models)	0	1-6	7-9	10
Comparison	0	1-2	3-5	5

Avoid the following datasets

- Fisher's Iris data
- The MPG car dataset
- Challenger data set